Name-Atharv Sawant

Matriculation number -100007048

Social Network Analysis Report

Dataset Information

The dataset used for this analysis was facebook_edges, analyzed using Gephi 0.11.1. The network represents a Facebook social network where nodes represent users and edges represent friendships or social connections.

Dataset/File Name: facebook_edges
Graph Type: Undirected Graph
Software Used: Gephi

Network Size

Before applying any filters, the network contained:

- Number of Nodes: 4,039
- Number of Edges: 88,374

This indicates a large and densely connected social network with many interactions between users.

Top 5 Nodes by Degree

Rank	Node ID	Degree
1	25	522
2	30	511
3	27	484
4	28	459
5	0	451

Top 5 Nodes by Betweenness Centrality

Rank	Node ID	Betweenness Centrality
1	25	0.025157
2	30	0.024168
3	27	0.021335
4	28	0.020175
5	0	0.019539

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Interpretation of the Network Structure

The Facebook network appears highly dense and strongly interconnected. The visualization shows that most users are connected through multiple pathways, which is common in real-world social media networks.

Community detection using modularity revealed several distinct groups within the network. Different modularity classes indicate clusters of users who are more closely connected to one another than to the rest of the network. This suggests the existence of friendship circles or communities based on shared interests or interactions.

The network also contains several hub nodes with very high degree values. These nodes are central to the network and significantly influence information flow. Additionally, nodes with high betweenness centrality act as bridges connecting different communities, making them important for maintaining overall network connectivity.

Overall, the network demonstrates key characteristics of social networks:

- Dense connectivity
- Community structure
- Influential hub nodes
- Strong clustering behavior

What I Learned

From this analysis, I learned how social network analysis can be used to understand relationships and interaction patterns in large datasets. Metrics such as degree and betweenness centrality help identify important and influential nodes within a network.

I also learned how Gephi can be used to visualize networks and detect communities through modularity analysis. The assignment improved my understanding of graph theory concepts and gave me practical experience in analyzing and interpreting social network structures.

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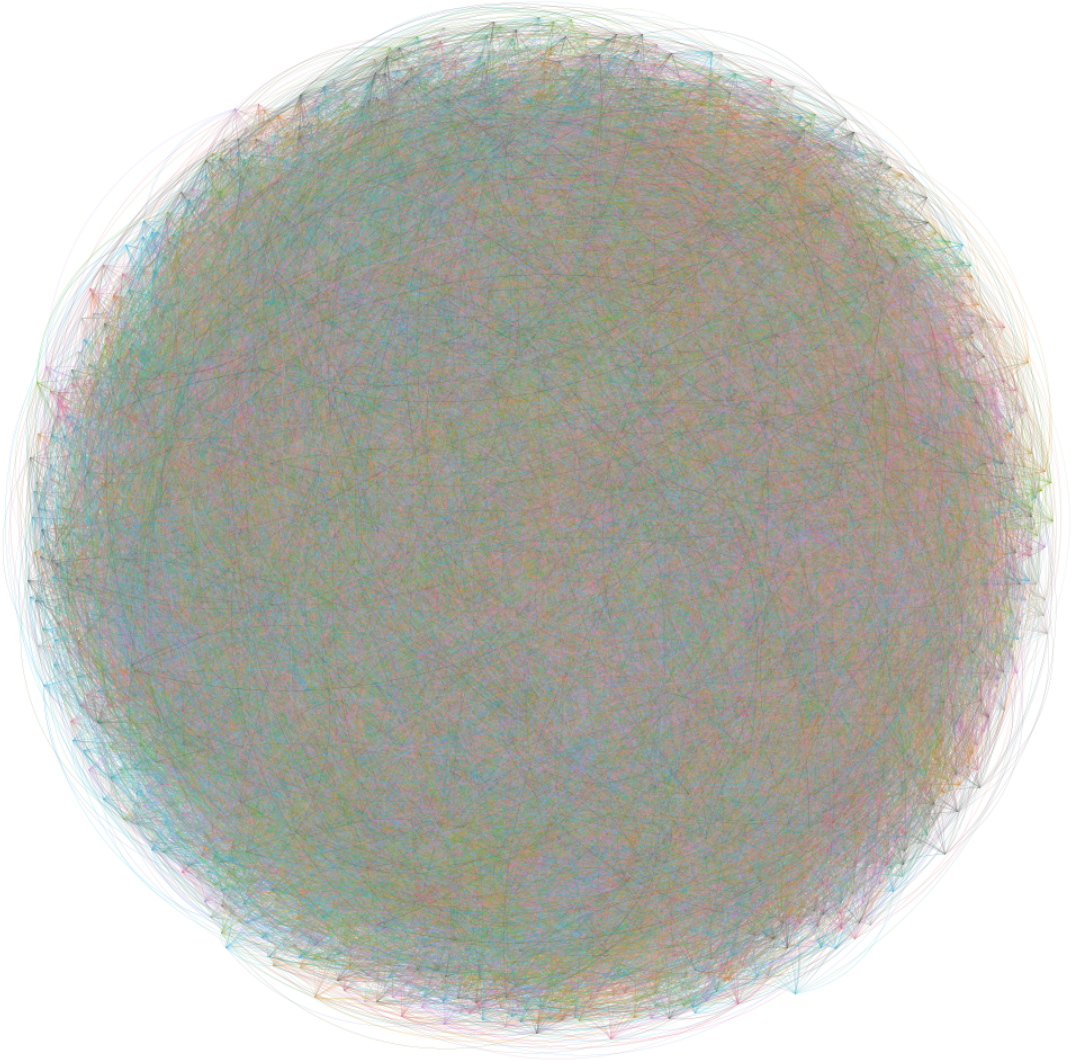

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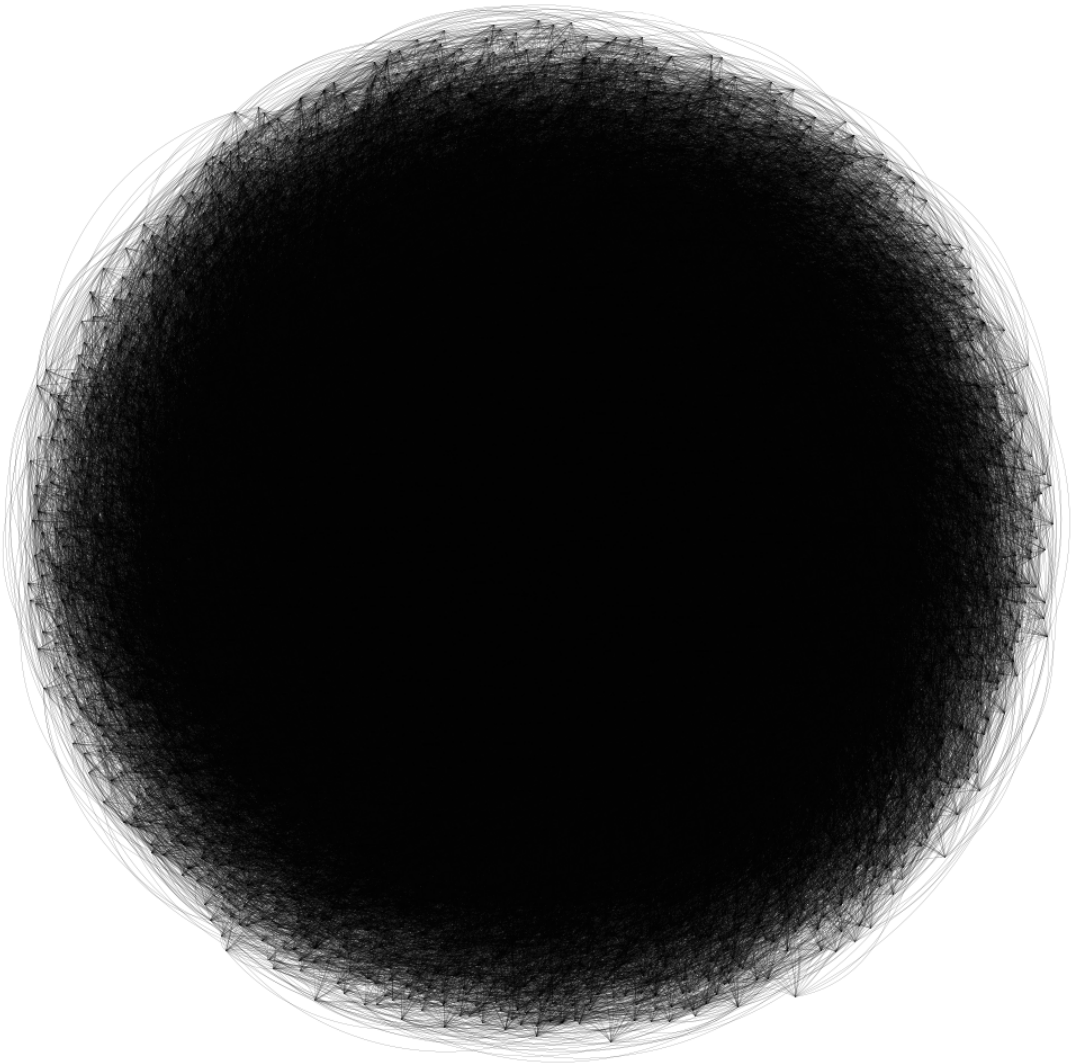
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